Supplementary Material: Themis: Exact Temporal Shortest Hyperpaths with Cycles for Pathway Inference in Cellular Reaction Networks

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The main paper states the structural theory of temporal feasibility, the ILP formulation, the cutting-plane algorithm Themis, and the main experimental results within its page limit. This supplement contains the full parameterized-complexity analysis (Sect. S1), the complete proofs of the main theorems (Sect. S2), and the supplementary experiments deferred from the main text (Sect. S3). Theorem, lemma, and corollary numbers written as "of the main text" refer to the numbering used in the main paper.

S1 Parameterized Complexity of T-SHP

We analyze T-SHP parameterized by five natural measures, situating the results relative to the static hyperpath and Steiner problems. Throughout, $n = |V|$, $m = |E|$, and $L = \sum_e d(e)$ denotes the total delay budget.

We first recall the two parameterized-complexity notions used below. A parameterized problem receives an instance of size n together with a parameter k. Fixed-parameter tractability (FPT) means a running time of $f(k)\, n^{O(1)}$ for a computable function f: once k is small, the problem is fast. The W-hierarchy $W[1] \subseteq W[2] \subseteq \cdots$ refines presumed intractability through parameterized reductions from weighted circuit satisfiability: a problem is W[t]-hard if every problem in W[t] reduces to it, and W[2]-hardness means that, unless the standard hypothesis FPT $\neq$ W[2] fails, no FPT algorithm exists. The canonical W[2]-complete problems are k-set cover and k-dominating set. Our reductions below are from k-set cover, so the W[2]-hardness of T-SHP (Theorem S1) inherits the presumed intractability of the entire class; the APX-hardness of the optimization variant follows from the inapproximability of set cover [18] through the approximation-preserving reduction below.

Theorem S1 (W-hardness by rank)**.** *The problem of deciding whether a temporal hyperpath of makespan at most M exists is W[2]-hard parameterized by the maximum tail size k, even when all delays are unit.*

Proof. Reduce from set cover with sets of size k, representing each set as a hyperedge whose tail is the chosen ground elements (available as sources) and whose head is a fresh "set" vertex; connect all set vertices to the target by unit-delay hyperedges. Any temporal hyperpath reaching the target must fire a collection of set hyperedges whose tails cover all ground elements required by the target's tail; the makespan is bounded by the number of fired layers, so a temporal hyperpath of makespan M exists iff a cover of size at most M exists. Since set cover is W[2]-hard parameterized by solution size and the reduction is parameter-preserving, the claim follows. The APX-hardness for bounded k follows from the fact that set cover remains APX-hard when every set has size at most 3 (see Böhmová et al. [20], proof of Theorem 1) together with the approximation-preserving reduction above. □

The W[2]-hardness anchors the problem in the same parameterized class as set cover and directed Steiner tree [23], and complements the static hardness results [3] and the enumeration lower bounds of Kurita and Mann [21]: even if one could enumerate candidate structures efficiently, which the main text shows is impossible in output-polynomial time for general directed hypergraphs, deciding temporal feasibility on top of them remains hard in the worst case.

Theorem S2 (FPT by length)**.** *T-SHP is fixed-parameter tractable parameterized by the hyperpath length ℓ.*

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Proof. Enumerate all subsets $F \subseteq E$ of size at most ℓ; there are $\binom{m}{\leq \ell} \leq m^\ell + 1$ of them. For each, check static reachability of t (linear-time closure [3]) and temporal feasibility by the cycle characterization of the main text (no positive-delay cycle, checkable in linear time by a strongly-connected-component (SCC) decomposition); if feasible, compute the makespan by the earliest schedule. Return the minimum over all feasible supports. The running time is $O(m^\ell \cdot (m+n))$, FPT in ℓ; this mirrors the FPT-by-length result of Casteigts et al. [6] for restless temporal paths and is essentially tight by Theorem S1. □

Theorem S3 (FPT by timed feedback vertex number)**.** *Let κ be the hypergraph timed feedback vertex number, the minimum number of vertex-appearances whose deletion makes every temporal hyperpath acyclic. Then T-SHP is FPT parameterized by κ.*

Proof. For a bounded feedback set W of timed appearances, branch over which hyperedges incident to W are selected (at most $2^{|W|}$ branches after standard reduction), remove W from the instance, and solve the residual acyclic problem by dynamic programming in topological order, accumulating delays. The result is FPT with running time $f(\kappa) \cdot \mathrm{poly}(n, m, L)$. The parameter generalizes the timed feedback vertex number introduced by Casteigts et al. [6] for temporal graphs to the hypergraph setting, and matches their observation that bounded waiting constraints, here positive delays, are the source of hardness, since the acyclic residual admits polynomial-time scheduling. □

Taken together, the results above give a fine-grained picture: Theorem S1 anchors W[2]-hardness by rank, Theorem S2 gives FPT by hyperpath length, and Theorem S3 extends FPT to the timed feedback vertex number; T-SHP is W[2]-hard in the "shape" parameters (rank, feedback set analogues) but FPT in the "solution-size" and "decomposition" parameters (length, treewidth, timed feedback vertex number), exactly mirroring the restless temporal path results of [6] and the static hyperpath results of [3, 21]. The integrality-ratio and inapproximability results [18, 19] further imply that, under standard complexity assumptions, no polynomial-time approximation can beat the static bounds, motivating the exact cutting-plane approach of the main paper.

S2 Complete Proofs of the Main Theorems

S2.1 Proof of Lemma 1 of the main text (minimal solutions suffice)

Proof. Let σ be a schedule witnessing makespan M for F, and let $C \subseteq F$ be the set of hyperedges that are not essential for reaching t. Iteratively remove any hyperedge whose removal leaves a set from which t remains reachable, until no further removal is possible, and call the resulting set P. By construction, P is a minimal set from which t is reachable, so P is a hyperpath. It remains to show that P admits a schedule of makespan at most M. Because $P \subseteq F$, the schedule σ restricted to P is a valid schedule for P, since every reactant of every $e \in P$ was available in F no later than $\sigma(e)$, and availability in the smaller set P can only improve, never worsen, the earliest firing time of any hyperedge of P when we recompute the canonical earliest schedule. The canonical earliest schedule of P therefore has makespan at most the makespan of σ restricted to P, which is at most M. □

S2.2 Proof of Theorem 1 of the main text (temporal cut theorem)

Proof. We prove the contrapositive. Let F be a hyperedge set with no hyperedge crossing the temporal cut of P, and let $C = \mathrm{cl}(F)$ be the temporal closure of F. We show $C \subseteq R \cup S$; since $t \notin R \cup S$ by assumption, this implies $t \notin C$, so F is not temporally feasible for t. The closure operator is monotone, so it suffices to show that every hyperedge of F that can fire within the closure has its head inside $R \cup S$. Indeed, consider any $e \in F$ with $T(e) \subseteq C$; by induction on the closure process, if $C \subseteq R \cup S$ before e fires, then $T(e) \subseteq R \cup S$, and since e does not cross the temporal cut, $h(e) \in R \cup S$; hence the closure remains inside $R \cup S$ after e fires. The induction starts with $S \subseteq R \cup S$ and closes over F, so $C = \mathrm{cl}(F) \subseteq R \cup S$. Now if F were a temporally feasible hyperpath for t, we would have $t \in \mathrm{cl}(F)$, contradicting $t \notin R \cup S$; hence no such F exists, and every temporally feasible hyperpath must contain a crossing hyperedge. Finally, no hyperedge of P crosses the cut by the definition of the temporal closure (every $e \in P$ with $T(e) \subseteq R \cup S$ has $h(e) \in R \cup S$, otherwise $h(e)$ would belong to $\mathrm{cl}(P)$), so the characteristic vector of P violates the cut inequality (1) of the main text. □

S2.3 Proof of Theorem 2 of the main text (cycle characterization of temporal feasibility)

Proof. For the forward direction, suppose F contains a directed cycle $C = (e_1, \ldots, e_k)$ with $h(e_i) \in T(e_{i+1})$ (indices modulo k) and $\sum_{i=1}^{k} d(e_i) > 0$. In any finite schedule σ for F, the schedule constraints of the main text force $\sigma(e_{i+1}) \geq \sigma(e_i) + d(e_i)$ along the cycle, and iterating around the cycle yields $\sigma(e_1) \geq \sigma(e_1) + \sum_i d(e_i) > \sigma(e_1)$, a contradiction. Hence no finite schedule exists and F is not temporally feasible. For the reverse direction, suppose every directed cycle of F has total delay zero. Since all delays are nonnegative, this means every hyperedge on every directed cycle of F has delay zero. Contract the strongly connected components of the dependency graph of F; within a component, all hyperedges have delay zero, so all vertices of the component can be produced simultaneously at the time the component is first entered; between components, the condensation is a directed acyclic graph, and we process it in topological order, accumulating delays along the way. Because t is statically reachable from S, this process reaches t in finite time, yielding a valid schedule. Thus F is temporally feasible. □

S2.4 Proofs of Corollaries 1 and 2 of the main text

Proof of Corollary 1 (soundness and completeness). Soundness is the temporal cut theorem of the main text. For completeness, suppose the cutting-plane algorithm terminates with F and $t \notin \mathrm{cl}(F)$; then the temporal cut of F is violated by F, contradicting termination. Hence $t \in \mathrm{cl}(F)$. □

Proof of Corollary 2 (degeneration to static cuts). With equal delays, availability order is the static dependency order, and a vertex is in $\mathrm{cl}(P)$ exactly when it is reachable in the static sense; the crossing condition $T(e) \subseteq R \cup S$, $h(e) \notin R \cup S$ is the static cut-crossing condition specialized to $R = \mathrm{cl}(P)$. The classical theorem (see Krieger and Kececioglu [5], Theorem 1) states that F is a superpath iff it crosses every cut, which is the degenerate form. □

S2.5 Proof of Lemma 2 of the main text (ILP correctness)

Proof. For the forward direction, let $F = \{e : x_e = 1\}$. The selection-consistency constraints ensure F is closed under the dependency relation, and the target-production inequality ensures t is produced. Let P be a hyperpath obtained from F by the trimming argument of Lemma 1 of the main text; we show P admits a schedule of makespan at most τ_t. The big-M constraints hold only for selected hyperedges, so we first argue that no selected hyperedge can lie on a positive-delay cycle: if $e_1, \ldots, e_k$ with $h(e_i) \in T(e_{i+1})$ (indices modulo k) and $\sum_i d(e_i) > 0$ existed, iterating the schedule inequalities around the cycle would force $\tau_{h(e_1)} \geq \tau_{h(e_1)} + \sum_i d(e_i)$, a contradiction since all τ are finite. Hence the support contains no positive-delay cycle, and by the temporal-distance theorem of the main text the earliest schedule of P is finite with makespan at most τ_t, since the schedule constraints and the target-production inequality are exactly its constraints. For the reverse direction, given a temporal hyperpath P with schedule σ, set $x_e = 1$ for $e \in P$, $x_e = 0$ otherwise, and τ_v to the earliest availability time of v under σ (with $\tau_s = 0$ for sources). The selection-consistency constraints hold by closure of P, the schedule inequalities hold by the schedule, and the target-production inequality holds because t is available at makespan $\sigma(t) = \tau_t$. □

S2.6 Proof of Theorem 4 of the main text (optimality of Themis)

Proof. Termination: each iteration adds at least one violated inequality. There are finitely many temporal cuts (at most 2^n) and finitely many cycles, so the process cannot continue forever; moreover, every added inequality is violated by the current solution and remains valid for all temporally feasible hyperpaths (the temporal cut theorem for cuts, the cycle characterization for cycle-elimination), so the restricted ILPs are never cut off from the optimum. Optimality: at termination, x^* satisfies every temporal cut, so by Corollary 1 of the main text the target lies in $\mathrm{cl}(F)$ (the temporal-closure oracle), and every cycle inequality, so by the cycle characterization the support contains a temporal hyperpath (the positive-delay-cycle oracle); by Lemma 2 of the main text, x^* encodes a temporal hyperpath of makespan τ_t. Since every restricted ILP is a relaxation of the full ILP, τ_t is a lower bound on the optimum; since x^* is feasible, τ_t equals it. □

Table S1: Temporal gain under synthetic bimodal versus measured-kinetics delays

Delay model	Gain instances	Fraction of instances	Gain range	Median gain
Synthetic bimodal	23	0.21% (23 of 11,116)	47–49	48
Measured k_{cat}	8	0.06% (8 of 14,401)	2–27	12.5

Table S2: Pathway recovery (F1 score) for static and temporal solutions over ten chain-structured Reactome pathways (17–48 reactions each) under the measured-kinetics delays; P^* is the annotated pathway's reaction set restricted to edges present in the database hypergraph, $|P^*|$ its size, and the F1 score of an inferred set X is $2|X \cap P^*|/(|X| + |P^*|)$; since inference is restricted to the pathway's induced sub-hypergraph, $X \subseteq P^*$ and the F1 score equals $2|X|/(|X|+|P^*|)$

Pathway	$\lvert P^*\rvert$	Static optimum	Temporal optimum
Dissolution of Fibrin Clot (R-HSA-75205)	18	1.0000	1.0000
Synthesis of Prostaglandins (PG) and Thromboxanes (TX) (R-HSA-2162123)	17	0.9697	0.9697
Iron uptake and transport (R-HSA-917937)	20	0.9744	0.9744
Interleukin-3, Interleukin-5 and GM-CSF signaling (R-HSA-512988)	25	1.0000	1.0000
Retinoid metabolism and transport (R-HSA-975634)	24	1.0000	1.0000
MHC class II antigen presentation (R-HSA-2132295)	26	1.0000	1.0000
Collagen degradation (R-HSA-1442490)	21	1.0000	1.0000
Nuclear signaling by ERBB4 (R-HSA-1251985)	34	1.0000	1.0000
Oxidative Stress Induced Senescence (R-HSA-2559580)	38	1.0000	1.0000
VEGFA-VEGFR2 Pathway (R-HSA-4420097)	48	1.0000	1.0000

Table S3: Cutting-plane iterations of Themis versus the full ILP on 24 representative real and synthetic instances: all 24 agree with the full ILP, 16 (67%) are solved at the initial ILP, and the hardest instances, requiring cycle-deadlock proofs, terminate within at most 3 iterations

Cutting-plane iterations	Instances	Fraction
0 (initial ILP)	16	66.7%
1	3	12.5%
2	4	16.7%
3	1	4.2%
Total	24	100%

S3 Supplementary Experiments

S3.1 Delay-model ablation

The delay model strongly affects the measured temporal gains: the synthetic bimodal distribution overestimates the number of gain instances by a factor of 2.9 (23 vs 8) and the gain range (47–49 vs 2–27), so measured kinetics are required for an honest estimate (Table S1).

S3.2 Pathway recovery (details of Sect. 4 of the main text)

Table S2 reports the per-pathway F1 scores summarized in the main text; the two semantics coincide on every pathway, so temporal planning does not distort biological content.

S3.3 Runtime and cutting-plane iterations

Themis is fast in practice. Over all 14,401 instances the median solve time is 3 milliseconds. In the cutting-plane iteration study, 16 of 24 representative real and synthetic instances (67%) are solved at the initial ILP without any cutting-plane iteration, and the hardest instances, requiring cycle-deadlock proofs, are certified within at most 3 iterations (Table S3). The full runtime distribution is shown in Figure S1, as an ECDF of solve times on a logarithmic time axis: for a database of n instances with solve times $t_1 \leq \cdots \leq t_n$, the ECDF is $\hat{F}(t) = \#\{i : t_i \leq t\}/n$, the fraction of instances solved within t seconds. We imposed a per-instance time budget of 300 seconds, far above every observed solve time; none was exceeded, and the largest observed solve time was 0.14 seconds.

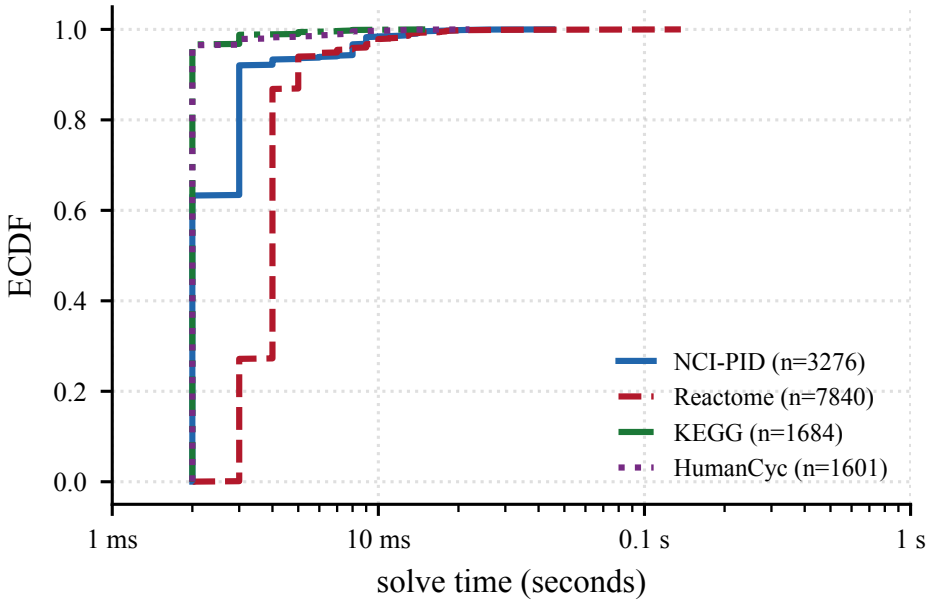


Figure S1: Runtime distribution over all instances as an ECDF on a logarithmic time axis. The median solve time is 3 milliseconds; the largest observed solve time is 0.14 seconds.

Table S4: KEGG instances where a cardinality-only static optimum is slow (measured kinetics)

Product	Cardinality-only makespan	Temporal makespan	Makespan gain
$b793bb05$; $ec32eef8$	24	1	23
$87cc5781$; $7a485584$	26	13	13
$3f2a8443$; $d0445887$; $f7317c3e$	29	16	13
$a42cd825$; $ec32eef8$	28	24	4
$bd3ec56a$; $ec32eef8$	25	23	2

S3.4 Exact linear verification (Theorem 3 of the main text)

PID, Reactome, KEGG, and HumanCyc are overwhelmingly single-product reactions, so the databases are B-hypergraphs and the exact temporal distance of the main-text theorem applies: the linear fixed-point computation of the main text returns $\delta(t)$ in sub-millisecond time, independently of the solver. Over all temporally feasible instances, the fixed point coincides with the ILP optimum on every single instance ($\delta(t) = T^*(t)$ on 13,693/13,693 = 100.0000% of them within a tolerance of 10^{-3}, which is far below one time unit since every makespan is integer-valued; the largest residual deviation is only 10^{-6}, a floating-point artifact of the LP solver). The infeasibility test agrees as well; all 706 instances declared temporally infeasible by the ILP (63 in PID, 639 in Reactome, 4 in KEGG, 0 in HumanCyc) satisfy $\delta(t) = \infty$ under the linear test, and so do the 2 statically infeasible instances, whose status the ILP left blank. The linear computation is about three orders of magnitude faster than the ILP (median 0.006 ms versus 3 ms per instance), so it doubles as an exact verifier of the solver's optimality, stronger evidence than the solver's own certificate, and as a practical pre-screen for the cutting-plane loop; the agreement is shown as Figure 4 of the main text.

S3.5 KEGG instances where a cardinality-only static optimum is slow

On top of the strict gains, the objective comparison shows that the *choice* of the static objective matters: on 5 KEGG instances the unique minimum-cardinality solution has makespan 24–29 while the temporal optimum completes in 1–24 time units (Table S4), even though the conservative tie-break (minimum total delay among cardinality-minimal solutions) already reaches the temporal optimum there (Table S4). A static planner that optimizes only cardinality can be arbitrarily slow in makespan terms even when a temporal gain is available; the tie-break decision, invisible to all static formulations, is what closes most of the remaining gap.

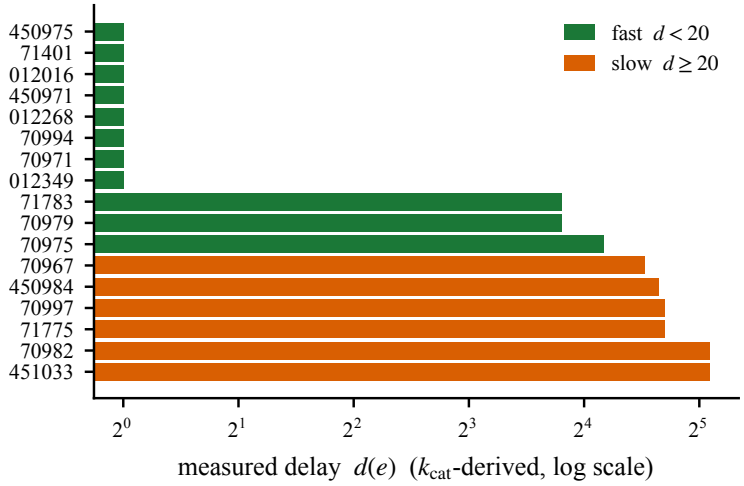


Figure S2: Measured delay $d(e)$ of the 17 member reactions of the TCA cycle (Reactome R-HSA-71403), a dimensionless k_{cat}-derived discretized latency on a log scale: 9 measured (14–34) and 8 default $d = 1$. Green: fast ($d < 20$, i.e. $k_{\text{cat}} \geq 10\, k_{\text{med}}$); orange: slow ($d \geq 20$). All delays are strictly positive, so no temporal schedule can traverse and return around this cycle (Theorem 2 of the main text).

S3.6 Real-world witnesses of the positive-delay-cycle characterization

Figure S2 shows the TCA case referenced in Sect. 4 of the main text.

S3.7 A representative closed positive-delay-cycle instance

The TCA case of the main text is a real-world witness of temporal infeasibility: a metabolic cycle whose member reactions are all reversible. Figure S3 shows the smallest exemplar of the same mechanism that we found among the 706 temporally infeasible instances, a two-step reversible pair that is closed in the full database hypergraph. The instance comes from the Reactome subset of Pathway Commons 12, with target 1-acyl glycerone 3-phosphate (GO3P, a precursor of phosphatidic acid). The static optimum selects three hyperedges: the two directions of the triose-phosphate isomerase pair *TPI1 dimer isomerizes DHAP to GA3P* (R-HSA-70454, glycolysis) and *TPI1 dimer isomerizes GA3P to DHAP* (R-HSA-70481, gluconeogenesis), both catalyzed by the same enzyme TPI1 (EC 5.3.1.1, GO:0004807), plus the downstream acylation *DHAP is converted to 1-acyl GO3P by GNPAT* (R-HSA-1483002, glycerophospholipid biosynthesis), which consumes DHAP.

The two TPI1 reactions form the positive-delay cycle DHAP $\leftrightarrow$ GA3P with total delay 40. In the full Reactome hypergraph neither DHAP nor GA3P has any producer outside this cycle, so the cycle is closed in the sense of the temporal-distance theorem of the main text; the static optimum therefore admits no finite schedule (cycle characterization), and the temporal ILP indeed returns *integer infeasible*. Both cycle edges carry measured kinetics in our k_{cat}-derived delay assignment ($d = 20$ in each direction), so the verdict does not depend on the delay model: any assignment of positive delays to the two reactions forces the contradiction $\sigma(\text{DHAP}) \geq \sigma(\text{DHAP}) + 40$.

The instance complements the TCA case at the opposite end of the size spectrum. TCA is a 17-step metabolic cycle; the TPI1 pair is the minimal reversible motif that already breaks any acyclic static plan, and the same motif recurs in the pathway databases because glycolysis and gluconeogenesis share the triose-phosphate isomerization step. Together the two cases show that temporal infeasibility is not an artifact of large synthetic cycles but a structural property of real metabolic and signaling networks.

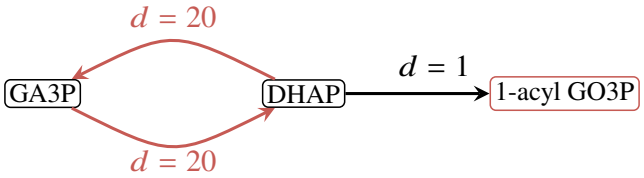


Figure S3: A minimal closed positive-delay-cycle instance from Reactome. The two directions of the triose-phosphate isomerase pair (R-HSA-70454 and R-HSA-70481, both with measured delay $d = 20$) form a closed cycle on DHAP $\leftrightarrow$ GA3P with total delay 40; the downstream acylation by GNPAT (R-HSA-1483002, $d = 1$) consumes DHAP to produce 1-acyl GO3P. The static optimum (three hyperedges) admits no finite schedule: any schedule would force $\sigma(\text{DHAP}) \geq \sigma(\text{DHAP}) + 40$

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